

California Polytechnic State University, San Luis Obispo

Project Report

Ultrasound-Based Tumor Segmentation
with Convolutional Neural Networks

Tenzin Konchok

Joseph Galicinao

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Dr. Jonathan Ventura

Introduction

For our final project, we implemented a deep learning system for breast ultrasound tumor segmentation. The goal of the project was to localize tumor regions at the pixel level rather than only classify whole images. This is an important medical imaging task because accurate tumor localization can support downstream analysis, treatment planning, and quantitative assessment. It is also a challenging task because ultrasound images are noisy, tumor boundaries can be ambiguous, and the number of tumor pixels is small relative to the large amount of background in each image. Accurate segmentation can also lead to more accurate feature extraction, allowing for more accurate classification. For the purposes of this project, classification and feature extraction is outside the scope of this project. Therefore, our main concern was segmentation.

Our approach used a U-Net based convolutional neural network trained on breast ultrasound images with corresponding segmentation masks. We began with a baseline U-Net architecture and then explored several modifications, including residual connections, attention gates, activation-function changes, and alternative loss functions designed to better handle class imbalance. We evaluated the models primarily using Dice coefficient and Intersection over Union (IoU), since these metrics are more informative for segmentation quality than pixel accuracy alone.

Background

Medical image segmentation has been strongly influenced by the U-Net architecture introduced by Ronneberger et al. U-Net is designed for dense prediction tasks and combines an encoder-decoder structure with skip connections that preserve spatial detail. This makes it especially suitable for biomedical segmentation problems where accurate boundaries matter.

Subsequent work has proposed several improvements to the original U-Net. Residual connections, inspired by ResNet-style architectures, can improve optimization by helping gradients flow more effectively through the network. Attention-based U-Net variants attempt to improve segmentation by suppressing irrelevant skip features and emphasizing regions that are likely to contain the target structure. In addition to architecture changes, many recent medical imaging studies emphasize that augmentation strategy and loss-function design can be just as important as the model structure itself, especially when data is limited and class imbalance is severe.

In this project, we used these ideas as the basis for an experimental comparison. We started from a standard U-Net and evaluated whether architectural changes or alternative loss functions improved segmentation quality on breast ultrasound data.

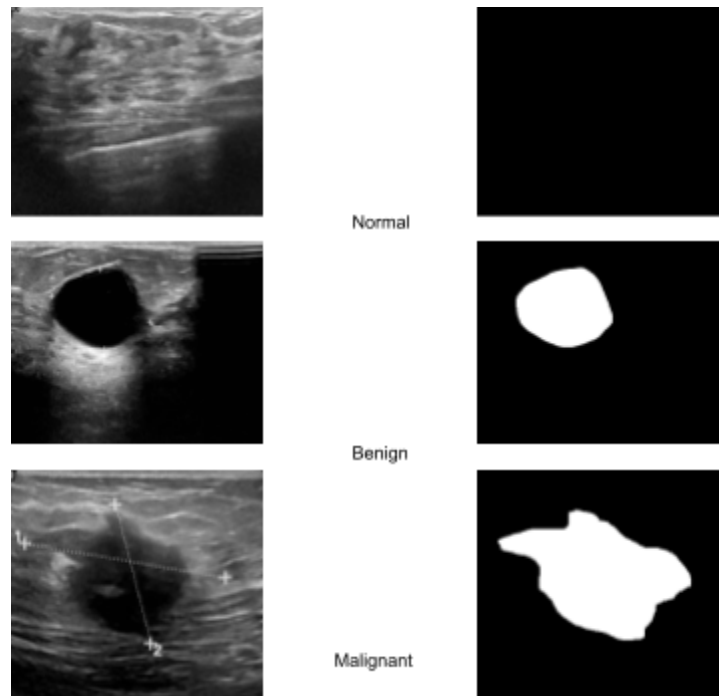
Dataset and Methods

We used the BUSI breast ultrasound dataset, which contains breast ultrasound images and associated segmentation masks. The dataset includes benign, malignant, and normal cases. In our implementation, images from all three categories were collected into a single pool, and their associated masks were paired through filename matching logic in the notebook.

Here is a link to the proposed dataset:

<https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset>

The notebook builds separate image-path and mask-path lists for benign, malignant, and normal folders, then concatenates them into a single dataset. A stratified 80/20 split is used so that the train and test partitions preserve the class proportions of the three categories. The figure below showcases all the corresponding image classes and their relative masks.



Preprocessing

All images and masks were resized to 128 x 128. Ultrasound images were loaded as grayscale images and normalized to the range [0, 1]. Ground-truth masks were loaded as grayscale mask images and converted into binary masks with values of 0 and 1. Using a fixed image size made it possible to batch the samples efficiently during training.

Augmentation

To improve generalization, the training pipeline applied simple geometric augmentation:

- Random horizontal flips
- Random vertical flips
- Random 90-degree rotations

These augmentations were applied to both the image and the mask together so that the segmentation target remained aligned with the input.

U-Net Model

The core model in the project is a U-Net style segmentation network. The encoder consists of repeated convolutional blocks followed by max pooling, while the decoder uses transposed convolutions to upsample intermediate features back to the original spatial resolution. Skip connections transfer encoder features to the corresponding decoder layers so that fine boundary information is preserved. The final layer uses a 1x1 convolution with sigmoid activation to predict a binary tumor mask.

U-Net Variations

We experimented with several model variations:

- Residual U-Net blocks
- Attention-gated skip connections
- Different activation functions such as ReLU, Leaky ReLU, and Swish

Loss Functions

We compared several training losses:

- Binary cross-entropy
- Dice loss
- IoU loss
- Binary focal cross-entropy with varying alpha values

This comparison was motivated by the strong foreground-background imbalance in segmentation. Since tumors occupy a relatively small portion of each image, focal-style and overlap-based losses may better emphasize difficult foreground pixels than plain binary cross-entropy.

Training Setup

Training was performed in TensorFlow using the Adam optimizer with an initial learning rate of $1e-4$. The notebook also used:

- Early stopping based on validation IoU

- Learning-rate reduction on plateau
- TensorFlow tf.data pipelines for batching and prefetching

In addition to the original architecture and loss experiments, we later ran a baseline batch-size sweep at 16, 32, and 64, followed by a combined experiment using BFC (alpha = 0.7) + Swish at the best-performing batch size.

Evaluation

We evaluated model performance using:

- Pixel accuracy
- IoU
- Dice

Although accuracy was recorded, Dice and IoU were the primary metrics of interest because segmentation datasets are dominated by background pixels. A high pixel accuracy alone does not guarantee good tumor localization. Below are the equations for IoU and dice scores.

$$\mathbf{IoU} = |\mathbf{P} \cap \mathbf{G}| / |\mathbf{P} \cup \mathbf{G}|$$

$$\mathbf{Dice} = 2|\mathbf{P} \cap \mathbf{G}| / (|\mathbf{P}| + |\mathbf{G}|)$$

Where:

$\mathbf{P}$ = predicted tumor mask
 $\mathbf{G}$ = ground-truth tumor mask

$|\mathbf{P} \cap \mathbf{G}|$ = number of overlapping tumor pixels

$|\mathbf{P} \cup \mathbf{G}|$ = number of pixels in either mask

Our evaluation compared the baseline U-Net against several revised versions of the architecture and against multiple loss-function choices. This allowed us to examine both architectural improvements and training-objective improvements.

Results

The first iteration of the project used a baseline U-Net and established a quantitative reference point for tumor segmentation performance. The baseline control setting achieved an accuracy of approximately 0.942, an IoU of 0.482, and a Dice score of 0.565. This showed that the model could learn meaningful tumor localization, but there was still clear room for improvement in overlap quality.

After evaluating the initial results, we explored a second round of improvements. These included residual connections, attention gates, activation-function changes, and alternative losses. The strongest reported result in our experiment table came from binary focal cross-entropy with $\alpha = 0.7$, which achieved an accuracy of 0.953, an IoU of 0.644, and a Dice score of 0.727. The α correlates to how much the model is punished if it misclassifies a foreground pixel. A larger α means more punishment. This result outperformed the baseline and suggests that handling class imbalance through the training objective had a major effect on segmentation quality.

Other experiments produced mixed results. A Swish activation variant also improved performance relative to the control. Attention and residual variants showed some benefits in certain metrics, but did not consistently outperform the best loss-function configuration. One focal-loss setting with $\alpha = 0.8$ performed poorly, indicating that aggressive class balancing can destabilize training or produce worse masks when poorly tuned.

To better understand training configuration effects, we ran an additional baseline batch-size sweep. The baseline model with batch size 16 achieved the strongest result in this sweep, with loss 0.1521, accuracy 0.9552, IoU 0.6449, and Dice 0.7151. In contrast, batch sizes 32 and 64 performed much worse, with IoU and Dice values near 0.17 to 0.18, and their qualitative predictions often collapsed toward mostly background masks.

We also tested a combined setting using BFC ($\alpha = 0.7$) + Swish at the best-performing batch size of 16. This combined model achieved loss 0.0204, accuracy 0.9476, IoU 0.6095, and Dice 0.6789. Although this was still a strong result, it did not outperform either the best original focal-loss result or the strongest baseline batch-size run.

Overall, the main insight from the evaluation is that better segmentation performance did not come only from adding architectural complexity. In this project, tuning the loss function appeared to be at least as important as changing the network structure.

Current Results Table

	Loss	Accuracy	IoU	Dice
Control: No residual connection, no attention, ReLU, Binary Cross Entropy Loss, Batch 128	0.217	0.942	0.482	0.565
Residual Connection	0.1378	0.951	0.244	0.661
Swish	0.289	0.949	0.618	0.692
Leaky ReLU	0.186	0.946	0.556	0.621
Attention	0.213	0.942	0.518	0.584
IoU Loss	0.683	0.947	0.558	0.621
Dice Loss	0.729	0.928	0.479	0.534
BFC (alpha = 0.6)	0.0267	0.9518	0.6113	0.688
BFC (alpha = 0.7)	0.023	0.953	0.644	0.727
BFC (alpha = 0.8)	0.14	0.394	0.099	0.164
Batch size = 16	0.1521	0.9552	0.6449	0.7151
Batch size = 32	0.6245	0.9094	0.1741	0.1741
Batch size = 64	0.6706	0.9094	0.1808	0.1808
BFC (alpha = 0.7) + Swish + (Batch size=16)	0.0204	0.9476	0.6095	0.6789

Outlook

At the start of the project, we expected that a baseline U-Net would provide a reasonable starting point and that at least one additional improvement cycle would produce measurable gains. Based on the current results, we were able to show that a revised version of the system improved substantially over the baseline, particularly when the training loss was tuned to account for class imbalance.

One of the most important lessons from this project was that segmentation should be evaluated with overlap metrics such as Dice and IoU rather than relying on accuracy. We also learned that architecture changes alone do not guarantee improvement. In our experiments, the choice of loss function played a major role in performance. This reinforced the idea that training strategy is often as important as network design in medical imaging tasks.

If we were to continue the project after the quarter ended, we would improve the evaluation protocol by using a clearly separated validation and final test set, perform more systematic hyperparameter tuning, and include more qualitative analysis of success and failure cases. Additional future work could include combining the best-performing architecture and loss settings, using more advanced augmentation, and testing whether the segmentation masks improve a downstream classification task.

Figures To Add Before Submission

1. Example ultrasound image with ground-truth mask
2. U-Net architecture diagram
3. Training/validation curve
4. Qualitative prediction examples
5. Final comparison chart or cleaned results table

References

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